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FYRE Pathway 1

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FYRE Reflection 1

In our first Fyre project we explored how circuits and how conductors work and developed our own moisture detection system. We began by learning the basics of what makes a circuit function and how they are made. After learning the basics, we toured a couple of different facilities beginning with Wilbur powerhouse where we used the sticker maker to make a masker for our circuit. The other facility was the clean room where we saw and learned about why the clean room is used but also necessary for the creation of chips due to the possibility of dust getting in the substance and completely ruining the circuit. For example, we were shown how photoresist could be applied over a disk using spin coating which is what we would end up using later ourselves.

Taking the sticker we created in Wilbur Powerhouse we placed it on glass insulator and spread silver on top. We then used a squeegee so that the silver was only in the small paths we cut into the sticker. After heating it on a hotplate for around a minute we removed the sticker resulting in a small electrode for us to experiment with. We then used the recently learned about strategy of spin coating to spread the PEDOTT over the electrode. This resulted in our electrode becoming a moisture detecting system. By measuring the resistivity of the circuit by the two ends, it would affect the conductivity of the PEDOT that was connecting the two halves of the electrode.

My group gained a lot from this experience because we ran into many difficulties that turned into learning opportunities as a result. We had a hard time taking off the sticker and applying it to the glass, so we had to innovate new ways to safely transfer the delicate material without it sticking to itself or ripping. Our solution was to slowly peel off multiple points instead of just one so that it would come off like one giant sheet this made it much easier to apply the sticker and help improve our teamwork. We were able to split up parts of the task so that each person would be better at one part so that we could make the electrodes faster working on our leadership and collaboration with one another. We also made sure to ask many questions so that we truly understood each part of the project so that we could really understand the lesson. As a whole. The teams cooperation and activity taught us not only how to be better teammates but also how the basics of electrodes and circuits work.